

Supplementary Material for: “OOD-Robust Degradation Modeling for Vehicular Battery Management Systems via Uncertainty-Driven Feature Manipulation with Topological Constraint”

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To complement the main paper, Table S1 details the comprehensive MAE and MSE performance of the SOH generalization experiments. These quantitative data support the corresponding visualizations presented in **Fig. 5 of the main paper** for a complete performance evaluation.

TABLE S1
COMPARISON EXPERIMENTAL RESULTS FOR THE OOD SOH ESTIMATION TASKS ON THE MATR, TONGJI, HUST, AND XJTU DATASETS.

Task	Task1		Task2		Task3		Task4		Task5		Task6
Metric	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
iTransformer [1]	5.9093	0.5257	1.1315	0.0350	1.3424	0.0317	1.0941	0.0218	2.7314	0.1079	21.3182
iTransformer* [1]	0.7803	0.0134	0.7618	0.0167	0.4686	0.0067	0.4534	0.0051	0.7615	0.0183	2.1108
PatchTST [2]	<u>1.1228</u>	0.0311	4.1024	<u>0.3776</u>	0.7981	0.0145	4.2731	0.4299	2.8899	0.1764	9.5076
PatchTST* [2]	0.6310	0.0098	0.6937	0.0136	0.4426	0.0071	0.4094	0.0046	0.8018	0.0183	3.6897
TSMixer [3]	16.2029	3.4749	2.5634	0.1033	5.1793	0.3115	2.9573	0.0980	3.1299	0.1231	35.9175
TSMixer* [3]	0.8600	0.0172	0.7558	0.0160	0.5797	0.0082	0.4818	0.0046	0.8081	0.0184	2.9360
PINN [4]	1.3596	0.0403	2.2216	0.0844	1.4187	0.0511	1.1438	0.0227	1.7547	0.0556	20.7705
PINN* [4]	0.5748	0.0061	1.3991	0.0361	1.3524	0.0269	0.7226	0.0107	1.0708	0.0207	8.9555
DR-Net [5]	4.2229	0.3324	2.6507	0.1112	3.4879	0.1749	3.6984	0.2049	3.4467	0.1925	16.0808
CRNN [6]	0.8260	0.0168	0.5589	0.0058	0.5295	0.0060	0.5508	0.0082	0.8301	0.0184	7.6085
MAGNet [7]	1.2410	<u>0.0244</u>	0.9305	0.0146	1.1954	0.0233	0.9216	0.0196	1.8908	0.0526	6.9453
DeNN-bagging [8]	1.7015	0.0658	2.1448	0.1194	2.1157	0.1141	2.1386	0.1330	2.8398	0.1719	10.3293
DeNN-fusion [8]	1.5901	0.0542	2.2918	0.1403	2.1127	0.1162	2.3360	0.1544	4.0800	0.2782	10.3575
DeNN-voting [8]	1.6199	0.0577	2.1824	0.1217	2.0499	0.1055	2.1529	0.1109	2.8789	0.1806	10.1680
MambaLithium [9]	1.2177	0.0338	<u>0.4801</u>	0.0038	<u>0.4002</u>	<u>0.0031</u>	<u>0.4162</u>	<u>0.0035</u>	<u>0.9160</u>	<u>0.0258</u>	<u>5.5906</u>
LSTM-DA [10]	3.0477	0.1140	2.8469	0.0945	3.1725	0.1164	3.4704	0.1360	3.4921	0.1370	12.8066
ConvMMD [11]	2.2734	0.0891	2.2941	0.0809	2.4317	0.0862	2.6367	0.0943	2.5047	0.0807	10.7025
DKTL [12]	2.2029	0.0889	2.4482	0.1134	2.6035	0.1153	2.8678	0.1272	2.7184	0.1135	10.5311
DDAN [13]	2.2708	0.0667	1.7973	0.0378	2.0361	0.0485	2.3470	0.0611	2.4232	0.0648	12.1931
RODTNet-LT** (ours)	1.1934	0.0484	0.9942	0.0367	0.9038	0.0330	2.4697	0.3593	3.1390	0.4989	9.6180
RODTNet-LT (ours)	1.0384	0.0208	0.4476	0.0030	0.3630	0.0020	0.2885	0.0014	0.4435	0.0032	4.0520

	Task6	Task7		Task8		Task9		Task10		Task11	
	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE
iTransformer [1]	8.5517	10.6146	1.6506	9.1256	1.5742	21.2702	8.0010	40.3827	20.7572	19.2927	4.6340
iTransformer* [1]	0.1204	8.8424	1.4328	1.3233	0.0277	2.9985	0.1737	2.8956	0.1635	2.8476	0.1570
PatchTST [2]	1.8668	16.9823	5.4530	9.2243	1.6822	14.7543	4.8261	48.1532	28.1815	15.8620	4.5693
PatchTST* [2]	0.5973	10.1355	1.8044	6.8517	0.9185	3.3279	0.1986	4.6602	0.3933	5.0112	0.4733
TSMixer [3]	15.7721	8.1766	1.0106	14.3296	2.9821	51.0444	26.6658	44.3269	22.9838	58.9715	35.8934
TSMixer* [3]	0.1595	9.3981	1.4881	1.6831	0.0443	2.7117	0.1365	4.0151	0.2957	3.7528	0.2954
PINN [4]	6.2435	25.2663	9.1038	5.9623	0.7837	7.9811	1.9312	22.6493	7.5184	13.4784	3.3358
PINN* [4]	2.2297	8.1476	1.3071	1.3870	0.0357	3.9588	0.4028	5.6329	0.9141	3.6541	0.3709
DR-Net [5]	3.9837	19.8591	5.6675	40.7034	21.2569	13.3021	3.0780	10.6609	2.0293	10.4743	1.7780
CRNN [6]	1.6485	4.5716	0.4770	3.3587	0.3612	5.8487	0.8072	11.0377	3.0051	6.9622	1.2084
MAGNet [7]	1.2605	<u>3.6287</u>	0.2369	3.2225	0.2692	<u>3.4654</u>	<u>0.2234</u>	7.5763	1.7185	4.1481	<u>0.4131</u>
DeNN-bagging [8]	1.9811	7.1896	0.8327	5.6580	0.7403	7.6599	1.1013	12.8670	3.4043	8.8430	1.6496
DeNN-fusion [8]	1.9799	8.5083	1.0813	5.9468	0.7503	8.3443	1.3646	12.5343	3.2910	8.9251	1.7611
DeNN-voting [8]	1.9101	7.4300	0.8838	5.5938	0.7390	7.7196	1.1244	12.7772	3.4203	8.9074	1.6948
MambaLithium [9]	<u>0.7132</u>	6.7626	0.8283	4.8486	0.6418	7.0512	1.2408	8.7161	2.1239	5.8056	0.8398
LSTM-DA [10]	2.8925	8.3498	1.0564	7.9172	1.3040	6.0889	0.6520	10.2645	2.3890	7.8039	1.1160
ConvMMD [11]	2.0985	7.8693	0.9537	6.7908	0.9026	5.9231	0.6199	11.1645	2.6587	7.6832	1.2462
DKTL [12]	2.1718	7.5353	0.8587	6.7470	0.9272	4.9570	0.5125	10.8147	2.5730	6.9723	1.1317
DDAN [13]	2.7491	8.1399	1.1031	7.3573	1.1792	5.9360	0.8296	9.0353	2.1172	5.9290	0.8442
RODTNet-LT** (ours)	2.0079	4.2455	0.4090	<u>1.7140</u>	<u>0.0635</u>	3.0794	0.2118	3.3261	0.2229	3.6471	0.8305
RODTNet-LT (ours)	0.3946	3.5394	<u>0.3020</u>	1.0781	0.0195	4.5385	0.3831	<u>5.0415</u>	<u>0.3875</u>	4.3266	0.3424

^a * indicates finetuning on the target domain.

^b ** indicates training directly on the target domain.

^c **Bold/underline** indicates the **best/second** best

^d Models utilizing pretraining-finetuning are excluded from comparison for fairness.

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